

Tutorato di Algoritmi e Laboratorio

Dottor Stefano Savino

email : stefano.savino@studium.unict.it

teams : STEFANO SAVINO

?

Struttura dei Tutorati

13 incontri per un totale di 25h

1h - 1,5h **Esercizi e Delucidazioni**

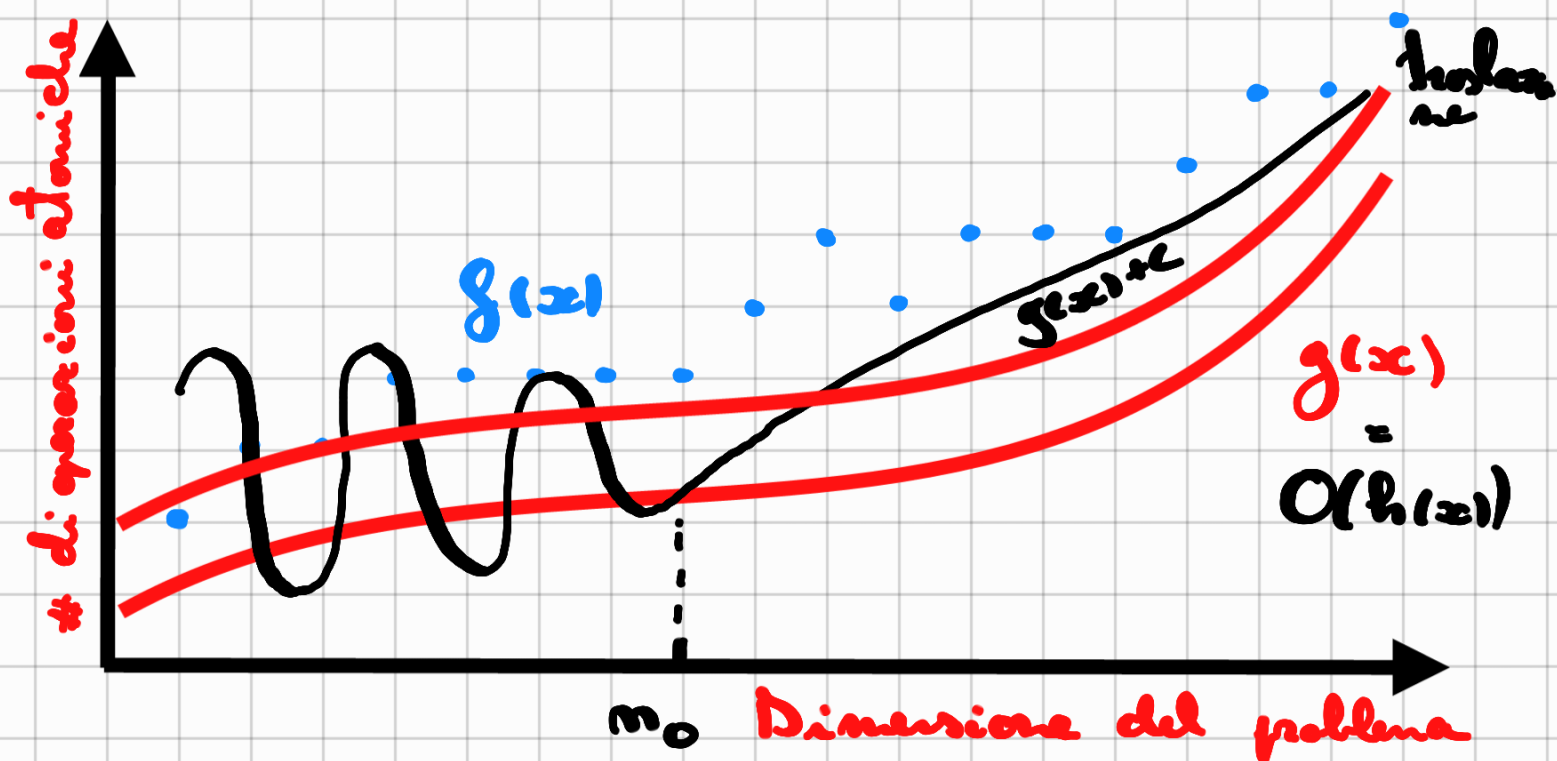
1h - 30m **Coding Contest**

Appunti caricati sul Team



questa si j
chiama moia

Una corsa contro al tempo



Quale tra blu e rosso rappresenta veramente il tempo di esecuzione di un algoritmo?

$$\exists m_0 \in \mathbb{R} \quad \exists c_1 \in \mathbb{R}^+$$

$$\exists c_2 \in \mathbb{R}^+ \quad c_1 g(x) \leq f(x) \leq c_2 g(x)$$

Definizioni?

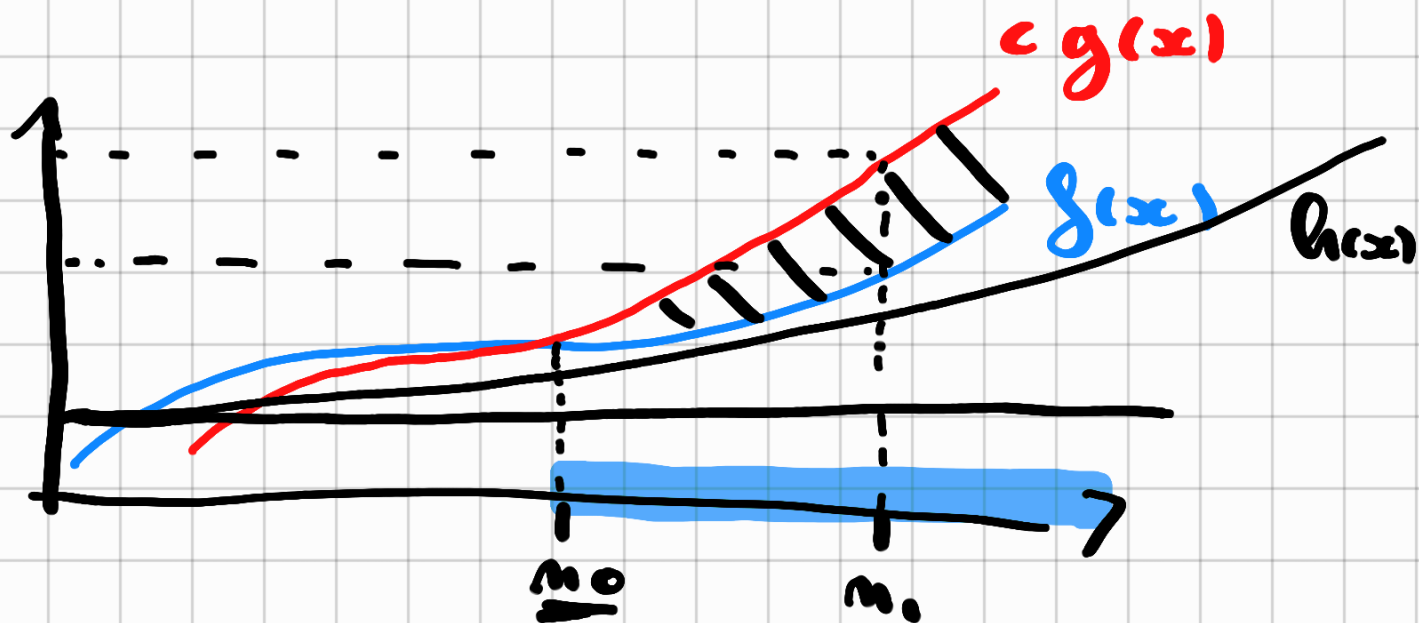
$$\underline{O} \quad \Theta \quad \underline{\Omega} \quad o \quad \omega$$

$$f(x) \sim O(g(x))$$

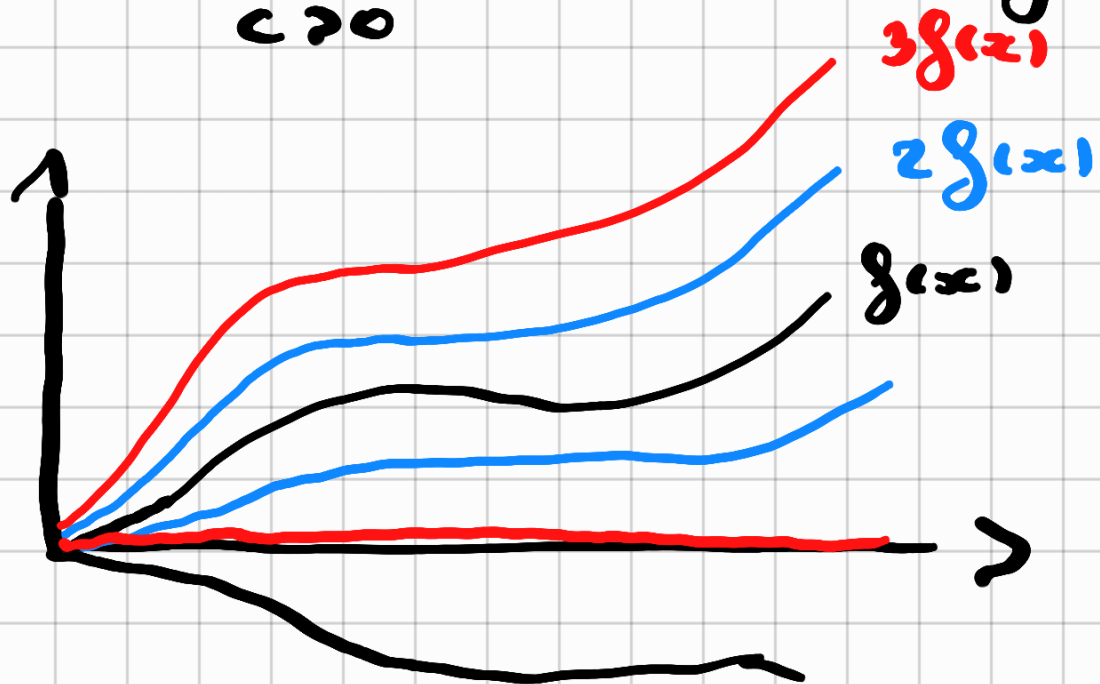
$$f(x) = o(g(x))$$

$$f(x) \in O(g(x))$$

$$\exists m_0 \in \mathbb{R} \quad \exists c \in \mathbb{R}$$



$$\exists m_0 \in \mathbb{R} \quad \exists c \in \mathbb{R}^+ : \forall m > m_0 \quad cg(m) \geq g(m)$$



$$x=2 \quad 5x+3 = 10+3 = 13$$

$$x=2 \quad \frac{1}{2}(5x+3) = \frac{13}{2} = 6,5$$

$$x=2 \quad 0(\dots) = 0$$

$$f(x) \in \Omega(g(x))$$

$$\exists m_0 \in \mathbb{R} \quad \exists c \in \mathbb{R}^+ : 0 < c \cdot g(x) \leq f(x)$$

Vediamo un po'...

$$n^2 + 2 \stackrel{?}{\in} O(n^3)$$

$$4n \log n + 7 \stackrel{?}{\in} O(n^4 \log n)$$

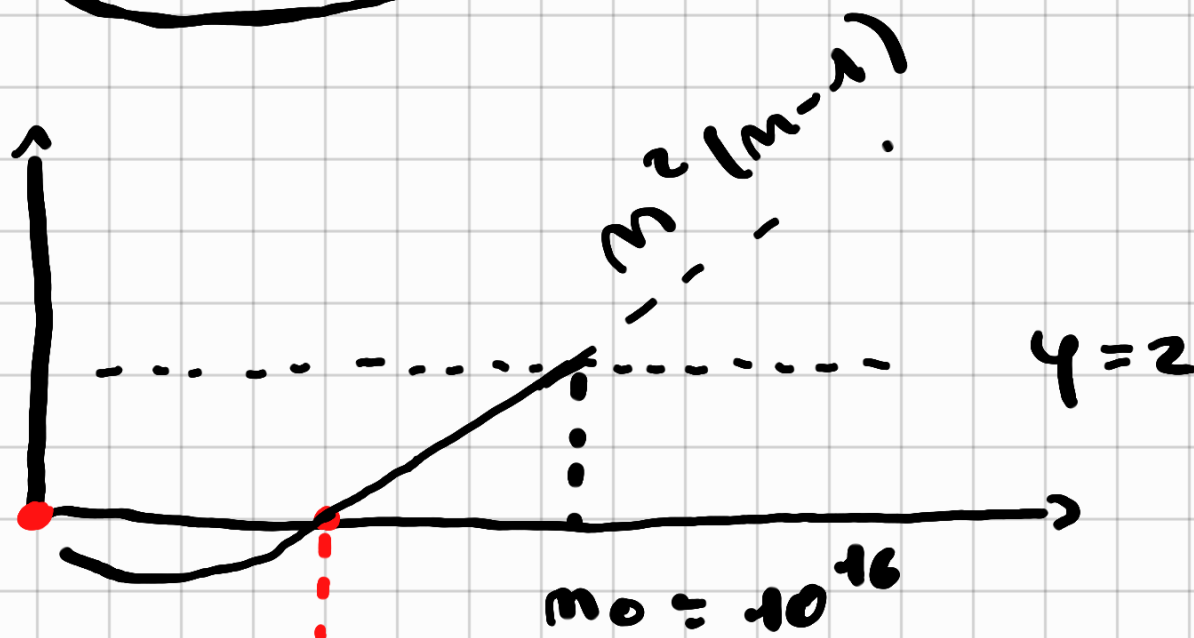
$$6n2^n \stackrel{?}{\in} \Theta(n^2 \log n + 100)$$

$$c = 1$$

$$n^3 \geq n^2 + 2$$

$$n^3 - n^2 \geq +2$$

$$n^2(n-1) \geq +2$$



Una corsa contro al tempo...

$$\begin{matrix}
 & \overbrace{\hspace{2cm}}^3 & \overbrace{\hspace{2cm}}^3 & \overbrace{\hspace{2cm}}^2 & \overbrace{\hspace{2cm}}^5 \\
 \underbrace{\hspace{1cm}}_5 \left\{ \begin{pmatrix} 7 & 7 & 7 \\ -2 & -1 & -2 \\ 3 & -4 & -4 \\ 2 & 2 & 2 \\ 5 & 7 & 9 \end{pmatrix} \right\} & \underbrace{\hspace{1cm}}_3 \left\{ \begin{pmatrix} 1 & 7 & 2 \\ 2 & 3 & 5 \\ 9 & 1 & 5 \end{pmatrix} \right\} & \underbrace{\hspace{1cm}}_2 \left\{ \begin{pmatrix} 6 & 9 \\ 7 & e \\ i & 12 \end{pmatrix} \right\} & \underbrace{\hspace{1cm}}_5 \left\{ \begin{pmatrix} 10 & 12 & 6 & 9 & 11 \\ 11 & 13 & 7 & 11 & 9 \end{pmatrix} \right\}
 \end{matrix}$$

$$\underbrace{\begin{pmatrix} a & b & c \\ d & e & f \end{pmatrix}}_3 \underbrace{\begin{pmatrix} x \\ y \\ z \end{pmatrix}}_1 = \begin{pmatrix} ax & by & cz \\ dx & ey & fz \end{pmatrix} \quad 2 \cdot 3 \cdot 1$$

P

5	3	3	2	5
---	---	---	---	---

	1	2	3	4	
1	0	45	48		
2		0	18	48	
3			0	30	
4				0	

chi e' A_{ij} ?

E' il costo

migliore per
 moltiplicare le
 matrici $A_i \dots A_j$
 $\left(\prod_{k=i}^j A_k \right)$

$$A_{1,3} = \min \left\{ \begin{array}{l} A_{1,2} + 5 \cdot 3 \cdot 2 \\ + A_{3,3} \end{array} , \begin{array}{l} A_{2,3} + 5 \cdot 3 \cdot 2 \\ + A_{1,1} \end{array} \right\}$$

$$A_{1,3} = \min \left\{ \begin{array}{l} A_{1,2} + A_{3,3} + 5 \cdot 3 \cdot 2 \\ \leftarrow A_{2,3} + A_{1,1} + 5 \cdot 3 \cdot 2 \end{array} \right\}$$

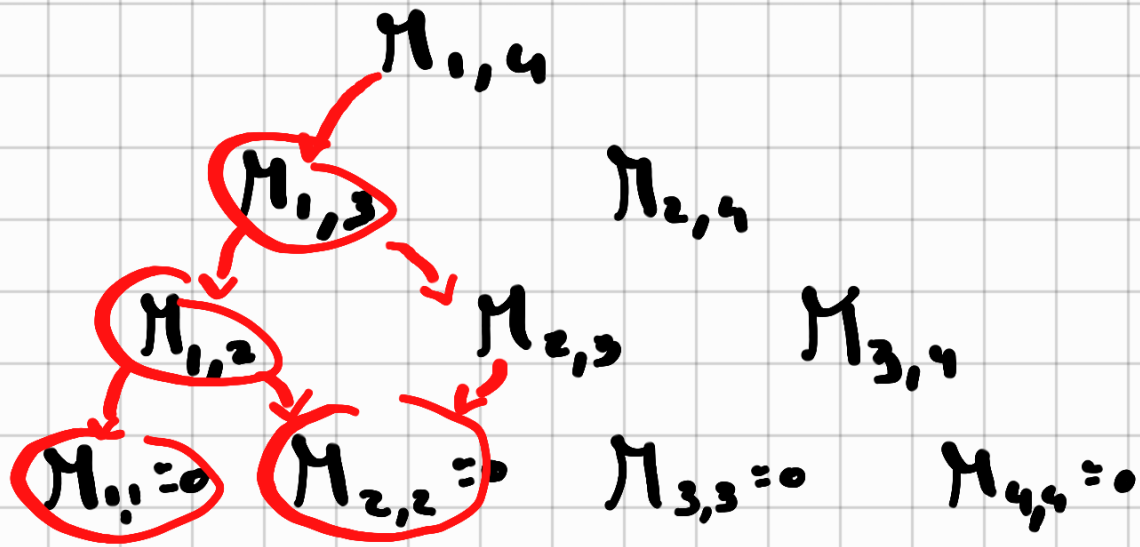
3-1 = 2

$$= \min \{ 75, 48 \} = 48$$

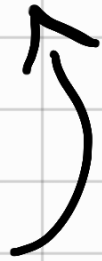
$$A_{2,4} = \min \left\{ \begin{array}{l} A_{2,3} + A_{4,4} + 3 \cdot 2 \cdot 5 \\ A_{2,2} + A_{3,4} + 3 \cdot 2 \cdot 5 \end{array} \right\}$$

$$= \min \{ 48, 60 \} = 48$$

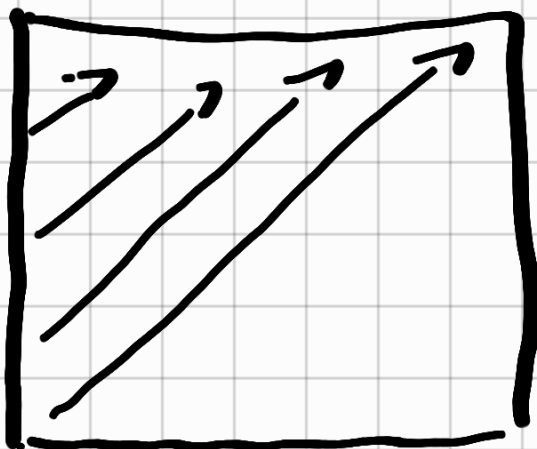
$$A_{1,4} = \min \left\{ \begin{array}{l} A_{1,1} + A_{2,4} + 5 \cdot 3 \cdot 5 \\ A_{1,2} + A_{3,4} + 5 \cdot 3 \cdot 5 \\ A_{1,3} + A_{4,4} + 5 \cdot 2 \cdot 5 \end{array} \right\}$$



Algorithmo ricorsivo con
 memorizzazione



programmazione
 dinamica



Esercizio

Moltiplicazione di matrici ...

Dato l'array P :

$\langle 7, 9, 2, 7, 3, 11 \rangle$

Dire :

- 1) Quante sono le matrici
- 2) Quali sono le loro dimensioni
- 3) Quanto costa moltiplicarle dalla prima alla quarta
- 4) Quanto costa moltiplicarle tutte quante
- 5) In quale indice della tabella è la soluzione migliore per moltiplicare dalla seconda alla quarta?